

1. Data Preparation and Cleaning (10 Marks)

- Import datasets into Power BI.
- Check data types.
- Remove duplicates.
- Handle missing values.
- Format StreamDate column.

Deliverable: Screenshot of cleaned data in Power Query Editor.

ID	123 StreamID		123 UserID		123 SongID		📅 StreamDate		123 ListeningTime		ABC DeviceType	
	📊		📊		📊		📊		📊		📊	
	Valid 100%		Valid 100%		Valid 100%		Valid 100%		Valid 100%		Valid 100%	
	Error 0%		Error 0%		Error 0%		Error 0%		Error 0%		Error 0%	
	Empty 0%		Empty 0%		Empty 0%		Empty 0%		Empty 0%		Empty 0%	
1	1		34		52		12-02-2025		139		Tablet	
2	2		108		11		19-07-2025		206		Mobile	
3	3		7		8		19-10-2025		251		Tablet	
4	4		146		4		23-11-2025		328		Desktop	
5	5		80		20		02-11-2025		330		Tablet	
6	6		6		43		16-03-2025		207		Tablet	
7	7		52		28		19-04-2025		304		Mobile	
8	8		109		5		20-03-2025		137		Mobile	
9	9		141		15		02-02-2025		279		Desktop	
10	10		103		37		11-08-2025		323		Tablet	
11	11		20		54		21-04-2025		95		Smart TV	
12	12		11		35		09-06-2025		65		Desktop	
13	13		173		3		21-03-2025		318		Mobile	
14	14		13		54		30-07-2025		291		Smart TV	
15	15		86		43		04-04-2025		257		Smart TV	
16	16		169		44		23-02-2025		94		Mobile	
17	17		56		10		03-12-2025		119		Tablet	
18	18		94		33		04-08-2025		184		Tablet	
19	19		106		55		05-04-2025		323		Mobile	
20	20		1		57		25-10-2025		123		Mobile	
21	21		117		59		02-12-2025		176		Smart TV	
22	22		8		24		01-06-2025		288		Tablet	
23	23		166		48		26-05-2025		325		Tablet	
24	24		51		11		21-01-2025		241		Mobile	
25	25		150		7		09-09-2025		240		Smart TV	

t ² ₃ SongID		A ^B _C SongName		A ^B _C Artist		A ^B _C Album		A ^B _C Genre		t ² ₃ Duration	
Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%
		1 Song_1		Arjun		Album_5		Pop			316
		2 Song_2		Neha		Album_5		Pop			334
		3 Song_3		Arjun		Album_7		Jazz			263
		4 Song_4		Vikram		Album_6		Indie			133
		5 Song_5		Riya		Album_3		Hip-Hop			263
		6 Song_6		Vikram		Album_12		Rock			161
		7 Song_7		Arjun		Album_15		Classical			318
		8 Song_8		Maya		Album_15		Jazz			216
		9 Song_9		Maya		Album_1		Hip-Hop			307
0		10 Song_10		Rahul		Album_2		Pop			143
1		11 Song_11		Riya		Album_3		Indie			352
2		12 Song_12		Arjun		Album_5		Classical			266
3		13 Song_13		Rahul		Album_5		Classical			334
4		14 Song_14		Arjun		Album_13		Hip-Hop			207
5		15 Song_15		Ananya		Album_13		Classical			172
6		16 Song_16		Surya		Album_11		Jazz			264
7		17 Song_17		Vikram		Album_1		Classical			348
8		18 Song_18		Ananya		Album_7		EDM			257
9		19 Song_19		Surya		Album_15		Indie			244
0		20 Song_20		Ananya		Album_11		Classical			215
1		21 Song_21		Ananya		Album_1		Indie			313
2		22 Song_22		Riya		Album_12		Indie			141
3		23 Song_23		Rahul		Album_11		Hip-Hop			163
4		24 Song_24		Vikram		Album_6		Jazz			220
5		25 Song_25		Maya		Album_12		Classical			219

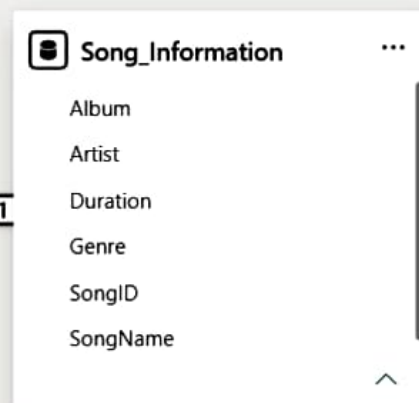
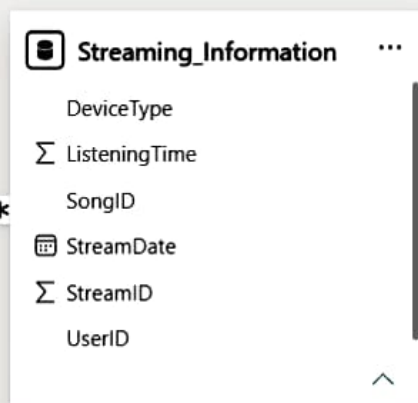
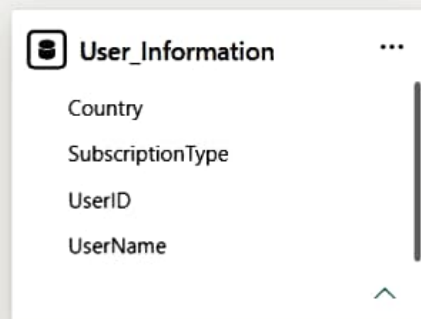
	1 ² 3 UserID		A ^B C UserName		A ^B C Country		A ^B C SubscriptionType	
	Valid	100%	Valid	100%	Valid	100%	Valid	100%
	Error	0%	Error	0%	Error	0%	Error	0%
	Empty	0%	Empty	0%	Empty	0%	Empty	0%
1			1	User_1		USA		Premium
2			2	User_2		UK		Premium
3			3	User_3		India		Free
4			4	User_4		USA		Free
5			5	User_5		India		Free
6			6	User_6		UK		Free
7			7	User_7		India		Premium
8			8	User_8		India		Premium
9			9	User_9		Australia		Free
10			10	User_10		Australia		Free
11			11	User_11		Canada		Premium
12			12	User_12		Germany		Premium
13			13	User_13		India		Premium
14			14	User_14		Germany		Premium
15			15	User_15		UK		Premium
16			16	User_16		USA		Free
17			17	User_17		Australia		Free
18			18	User_18		Australia		Premium
19			19	User_19		UK		Free
20			20	User_20		Germany		Free
21			21	User_21		UK		Premium
22			22	User_22		UK		Premium
23			23	User_23		UK		Premium
24			24	User_24		USA		Premium
25			25	User_25		UK		Free

Deliverable: Screenshot of cleaned data in Power Query Editor.

2. Data Modeling (10 Marks)

- Create relationships using SongID and UserID.
- Configure cardinality appropriately.

Deliverable: Screenshot of Data Model View.



3. Enhanced Data Insights (10 Marks)

- Create Conditional Column: ListeningTime $\geq$ 200 $\rightarrow$ Highly Engaged
- Use Column From Examples to create Artist Initial.

Deliverable: Screenshot of transformed data.

4. Calculations Using DAX (10 Marks)

Engagement		DeviceType			
StreamID	UserID	SongID	StreamDate	ListeningTime	DeviceType
1	34	52	12 February 2025	139	Tablet
2	108	11	19 July 2025	206	Mobile
3	7	8	19 October 2025	251	Tablet
4	146	4	23 November 2025	328	Desktop
5	80	20	02 November 2025	330	Tablet
6	6	43	16 March 2025	207	Tablet
7	52	28	19 April 2025	304	Mobile
8	109	5	20 March 2025	137	Mobile
9	141	15	02 February 2025	279	Desktop
10	103	37	11 August 2025	323	Tablet
11	20	54	21 April 2025	95	Smart TV
12	11	35	09 June 2025	65	Desktop
13	173	3	21 March 2025	318	Mobile
14	13	54	30 July 2025	291	Smart TV
15	86	43	04 April 2025	257	Smart TV
16	169	44	23 February 2025	94	Mobile
17	56	10	03 December 2025	119	Tablet
18	94	33	04 August 2025	184	Tablet
19	106	55	05 April 2025	323	Mobile
20	1	57	25 October 2025	123	Mobile
21	117	59	02 December 2025	176	Smart TV
22	8	24	01 June 2025	288	Tablet
23	166	48	26 May 2025	325	Tablet
24	51	11	21 January 2025	241	Mobile
25	150	7	09 September 2025	240	Smart TV
26	170	12	23 February 2025	87	Desktop
27	147	25	11 March 2025	309	Mobile
28	125	60	27 September 2025	160	Smart TV
Streaming Information (200 rows) Column: Engagement (2 distinct values)					

4. Calculations Using DAX (10 Marks)

Create measures for:

- Total Streams
- Total Listening Time
- Average Listening Time
- Premium Users
- Total Songs

Deliverable: Screenshot of DAX measures and results.

200

Total Streams

40K

Total Listening Time

200.45

Average Listening Time

87

Premium Users

60

Total Songs

5. Visualization and Interactive Features

Create visuals:

- Streams by Genre
- Top Artists by Streams
- Listening Time by Country
- Subscription Type Distribution
- Device Usage Analysis
- Monthly Streaming Trend

Interactive Features:

- Genre Slicer
- Artist Slicer
- Country Slicer
- Subscription Type Slicer

Genre

☐ Classical

☐ EDM

☐ Hip-Hop

☐ Indie

☐ Jazz

☐ Pop

☐ Rock

Artist

☐ Ananya

☐ Arjun

☐ Asha

☐ Karan

☐ Maya

☐ Neha

☐ Rahul

Country

☐ Australia

☐ Canada

☐ Germany

☐ India

☐ UK

☐ USA

SubscriptionType

☐ Free

☐ Premium

60

Total Songs

Total Streams by Genre

Total Streams by Year, Quarter, Month and Day

87

Premium Users

Count of UserID by SubscriptionType

200.45

Average Listening Time

Total Streams by Artist

200

Total Streams

Total Streams by DeviceType

40K

Total Listening Time

Total Listening Time by Country

6. Final Dashboard and Power BI Service (20 Marks)

Dashboard should include:

- Total Streams KPI
- Total Listening Time KPI
- Average Listening Time KPI
- Premium Users KPI
- Total Songs KPI
- Streams by Genre
- Top Artists by Streams
- Listening Time by Country
- Device Usage Analysis
- Monthly Streaming Trend

